Kolitha Warnakulasooriya

Curriculum Vitae

SUMMARY

Results-driven Researcher & Software Engineer with 6+ years of experience driving enterprise software systems, AI/ML innovation across Robotics/UAV systems, and real-time systems, and currently working as an Assistant Professor at the Department of Electrical and Computer Engineering. My current research track focuses on developing and identifying methods to neutralize and countermeasures for Drone/UAV or multiple UAV Threats. My proven track record includes engineering a novel forest-inspired swarm intelligence optimization algorithm that outperformed 10 state-of-the-art algorithms by 67% across 62 benchmark tests, developing a distributed leader selection algorithm for heterogeneous UAV swarms controlling 15 agents in formation, and managing three UAV research projects that increased completion rates by 25%.

Throughout my career at the University of South Alabama and SyscoLabs Technologies, I've consistently demonstrated expertise in experimental design, system architecture, and scalable cloud solutions while ensuring high-quality standards and robust performance. I've implemented machine learning techniques and custom loss functions in PyTorch, engineered heterogeneous UAV coordination frameworks in Java, and developed multi-agent simulation platforms embedding 26 algorithms and 76 benchmark problems.

I excel in high-stakes environments where I can leverage my research leadership, analytical thinking, and technical expertise to deliver measurable results. With a PhD in Computing (Computer Science) and extensive experience spanning industry and academia, I combine theoretical rigor with practical implementation skills to create innovative solutions that drive substantial technological advancement.

Selected Accomplishments

- Applied deep learning techniques and custom loss functions in PyTorch to refine forest-inspired
- swarm optimization models, conducting 14 extensive experiments and 62 benchmark tests to validate 67% performance gains over leading algorithms.
- Engineered a heterogeneous UAV coordination framework in Java, enabling real-time control of up to 15 agents (Parrot, Anafi, Tello) using distributed leader-election protocols for threat neutralization in 42+ live trials.
- Developed a multi-agent swarm intelligence simulator embedding 26 algorithms and 76 benchmark scenarios, featuring a custom GUI and automated data-set handling for reproducible testing.
- Led the restructuring of a BFF microservice, implementing extensive unit and integration tests to achieve over 90% code coverage and integrating Docker/Kubernetes for seamless deployments.

RESEARCH INTERESTS

Artificial Intelligence, Distributed Computing Systems, Edge Computing, Embedded Systems, Swarm Intelligence, Intelligent Robotics, UAV Systems, Robot Controls, Aerial Robotics, Swarm Robotics

ABOUT

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ORCID: orcid.org/0000-0002-4619-7027
Google Scholar: scholar.google.com/citations?user=r1q4ADUAAAAJ&hl=en
Semantic Scholar: www.semanticscholar.org/author/Kolitha-Warnakulasooriya/24346996

ACADEMIC QUALIFICATIONS

Ph.D. in computing **Jan 2022 - May 2025**
University of South Alabama, USA
GPA: 3.91 Overall
Dissertation: Choosing Robust Leadership: Encompassing the Best-of-N Model and Swarm Intelligence Optimization for Heterogeneous Multiple Autonomous Unmanned Aerial Vehicle Systems
https://www.proquest.com/docview/3190089835

This dissertation propose a new, distributed, and autonomous leader selection algorithm for large unmanned aerial vehicle (UAV) systems, designed to address the limitations of hierarchical control structures and leader failure. The proposed methodology is rooted in swarm intelligence and formalized within the best-of-n model. It included a new swarm intelligence algorithm called Tree Search Optimization Algorithm (TSOA), which significantly outperforms existing methods. The TSOA algorithm is used to develop a highly accurate and efficient leader selection process, triggered by an automatic fallen UAV capture method. This dissertation ultimately enhances the reliability and safety of large UAV applications by providing a robust, autonomous, and efficient solution for leader selection and task management.

Artificial Intelligence, Advanced Artificial Intelligence (A)
Advanced AI Theory and Programming (A)
Advanced Big Data Systems (A)
Cybersecurity, Advanced Cybersecurity (A)
Cyber Physical Systems (A)
Computer Architecture (A)
Digital Investigation (A)
Computer Echosystems and Cloud Infrastructures (A)

B.Sc. (Hons) in Information Technology **Mar 2014 - July 2018**
University of Moratuwa, Sri Lanka
GPA: 3.55 Second Class Upper Division

Robotics (A)
Embedded Systems (A)
Computer & Network Security (A)
Artificial Intelligence (A)

Thesis Project: Cucumber plucking robot for greenhouses in Japan
Contribution: Developed the mathematical method to predict and identify the best plucking point by using the Fourier analysis of color patterns spread on the fruit.

Embedded Systems Design & Development using Microcontroller Programming **Dec 2013**
University of Colombo, Sri Lanka

EMPLOYMENT EXPERIENCE

Assistant Professor **Aug 2025 - Present**
Department of Electrical and Computer Engineering, University of South Alabama, United States

Conducting research related to intelligent UAV systems, safety measurements from a UAV threat, and teaching related to computer engineering subjects. Collaborating with research teams from other faculties.

- Teaching
    - 400/500 – Digital Computer Architecture | Fall
    - 200 – Digital Logic Design | Fall/Spring
    - 400/500 – Introduction to Robotics | Spring
    - 300 – Microprocessor System Interfacing
- Conducting research related to improving national security for autonomous robot/UAV systems and navigation, developing models, methods, and equipment for acquiring actions.
- Student advising for 16+ students, and Core technical advisor for senior design projects (Undergraduate) regarding UAV systems and robotic systems.
- Advising a Master's of Electrical Engineering student as a thesis committee member.
- Faculty advisor for the IEEE HKN chapter and the Robotics Club.

Research Assistant **Jan 2022 – May 2025**
School of Computing, University of South Alabama, United States
Leads the design and implementation of cutting-edge AI and swarm intelligence projects within the Department of Computer Science, collaborating with cross-functional teams to translate theoretical concepts into production-ready research systems. Manages all phases of the research lifecycle, from literature reviews and experimental design to simulator development and UAV field trials, ensuring rigorous scientific methodology and reproducible results.

Machine Learning & Optimization

- Applied deep learning techniques and custom loss functions in PyTorch to refine the forest-inspired swarm optimization model, conducting 14 extensive experiments and 62 benchmark tests to validate performance gains.
- Tuned hyperparameters through automated pipelines, leveraging TensorBoard and custom logging to achieve a 67% improvement in 10+ leading algorithms in convergence speed and resource utilization.
- Authored research papers detailing algorithm architecture, experimental protocols, and results, submitting them to top-tier evolutionary intelligence journals and conferences.

Simulation & Evaluation

- Developed a multi-agent swarm intelligence simulator embedding 26 algorithms and 76 benchmark scenarios, featuring a custom GUI and automated data-set handling for reproducible testing.
- Automated large-scale testing through Python scripts, collecting metrics on convergence, stability, and computational load across variable problem dimensions.

UAV System Architecture

- Engineered a heterogeneous UAV coordination framework in Java, enabling real-time control of up to 15 agents (Parrot, Anafi, Tello) using distributed leader-election protocols.
- Integrated real-time sensor data streams and robust communication channels to support formation control and social threat neutralization in 42+ live trials.
- Designed modular API endpoints and containerized microservices (Docker, Kubernetes) for scalable deployment and field integration.

Lead Software Engineer (Associate) **Aug 2021 – Jan 2022**

SyscoLabs Technologies, Colombo, Sri Lanka

Served as Epic Owner for the Sysco Shop CRM application, overseeing feature planning, sprint execution, and post-release performance analysis. Fostered collaboration across product, QA, and DevOps teams by applying Scrum best practices and maintaining comprehensive Confluence documentation.

- Guided a team of 10+ engineers to deliver six critical Epics, employing system design principles and thorough code reviews to maintain high-quality standards.
- Streamlined Agile processes by crafting detailed sprint roadmaps, facilitating daily stand-ups, and curating JIRA dashboards, resulting in a 50% reduction in administrative overhead.
- Acted as the first technical responder for production incidents, coordinating cross-functional troubleshooting efforts to achieve zero downtime for key business services.

Senior Software Engineer **Dec 2019 – Aug 2021**

SyscoLabs Technologies, Colombo, Sri Lanka

Maintained and enhanced large-scale eCommerce platforms, utilizing software engineering practices, working across front-end systems (React, Redux) and back-end microservices (Spring Boot, Node.js) within AWS EC2 containers. Drove improvements in predictive analytics, test automation, and CI/CD pipelines while mentoring junior engineers in best practices.

- Led the restructuring of the BFF microservice, implementing extensive unit and integration tests to achieve over 90% code coverage and integrating Docker/Kubernetes for seamless deployments through Infrastructure as Code tools.
- Conducted in-depth code reviews, troubleshooting critical issues, and provided customer support and leadership in on-call rotations, ensuring continuous availability and reliability.
- Enhanced arrival time estimation models by 15–20% through statistical analysis and algorithmic refinements in Python.
- Leveraged software engineering skills to design and develop the Sysco Shop e-commerce web application and the Sysco Delivery mobile app.
- Proficiently troubleshooted complex and straightforward issues and contributed as an on-call representative for production incidents, achieving zero production incidents reported related to responsibility components.
- Enhanced the application component features through precise development plans, utilizing best software engineering practices, ensuring testability and quality of functionality via unit tests, component tests, integration tests, regression tests, E2E testing, and sea trials, fixing bugs, and enhancing overall performance efficiency.
- Improved system reliability by monitoring the overall system with Dynatrace and Datadog platforms.
- Offered guidance and mentorship to junior engineers while providing technical and cultural support during the onboarding process for new hires.

Software Engineer **Jul 2018 – Nov 2019**

SyscoLabs Technologies, Colombo, Sri Lanka

Developed responsive web and mobile applications using React, React Native, and ES6, emphasizing object-oriented design and state management patterns. Implemented RESTful, GraphQL, and WebSocket APIs within a microservice ecosystem, incorporating AWS Lambda and API Gateway for high availability.

- Engineered scalable UI components and backend services (Spring Boot, Express.js) within AWS Cloud Infrastructure, and developed an open-source in-memory mock service for ML testing, accelerating experimental workflows by 40% and ensuring high availability across front-end and microservice architectures.
- Utilized software engineering techniques to develop new features for the Sysco Shop web application and the Sysco Delivery mobile app.
- Created clear, well-structured, and maintainable new React and React Native components, implementing effective state management with Redux and Recoil frameworks while adhering to test-driven development principles.
- Implemented RESTful and GraphQL APIs in a microservice-based architecture using Spring Boot and Node.js Express frameworks.
- I ensured the testability of the software components through unit testing, component testing, integration testing, E2E testing, and sea trials.
- Collaboratively developed a machine learning strategy to enhance the application's customer experience and created a high-performance mockup API generator that eliminates the need for third-party mock services.

Associate Software Engineer **Dec 2016 – July 2017**
99X Technology, Colombo, Sri Lanka
Developed and fixed an ASP.NET application for Uvdal booking systems, incorporating HTML/CSS for the booking calendar. Gained experience as a full-stack developer within the .NET Framework, demonstrating flexibility and collaboration skills.

Research Assistant **Oct 2016 - Jan 2018**
4ONSE – Switzerland National Science Foundation
SUPSI University, Switzerland, and University of Moratuwa, Colombo, Sri Lanka

- Developed a low-cost IoT-based weather station network, incorporating robust data acquisition and transmission protocols.
- Designed double-layered PCB boards using Proteus and ORCAD.
- Implemented data processing algorithms using C++, achieving real-time weather data analysis.
- Deployed a central data storage service for storing and retrieving weather data, leveraging SQL for efficient querying.
- Achieved 99% data accuracy through rigorous quality control and programming, demonstrating data analysis and system integration proficiency.

RESEARCH PUBLICATIONS (H-INDEX 5, CITATION RECORD 81)

K. Warnakulasooriya, A. Segev, “Tree Search Optimization Algorithm: A novel heuristic optimizer based on knowledge processing in biological trees,” 2026, Under submission.

K. Warnakulasooriya, A. Segev, “Identifying Crashed or Disconnected Agents in Multi UAV Consensus Network,” 2026, Under submission.

K. Warnakulasooriya, H.E. Sevil, A. Segev, “Cooperative Optimized Leader Selection in Large UAV Swarms by Maximizing the Average Controllability in a Hierarchical Control Structure,” 2026, AIAA SciTech.

K. Warnakulasooriya, A. Segev, “Comparative analysis of accuracy and computational complexity across 21 swarm intelligence algorithms,” *Evolutionary Intelligence*, 2025.

K. Warnakulasooriya, and A. Segev, “Attacks, Detection, and Prevention on Commercial Drones: A Review,” *International Conference on Image Processing and Robotics*, 2024.

K. Warnakulasooriya, R. G. Benton, and A. Segev, “Priority Basis Task Allocation for Drone Swarms,” *AAAI Association for the Advancement of Artificial Intelligence,* 2023.

B. H. Sudantha, K.Warnakulasooriya and Y. P. Jayasuriya and G. R. Ratnayaka and P. K. S. Mahanama and E. J. Warusavitharana and S. Weerasinghe, “Open-Source Implementation of an Integrated, Low-Cost Environmental Monitoring System (EMS) for developing countries,” *Bhumi The Planning Research Journal*, 2018.

K. Warnakulasooriya, C. Premachandra, B. H. Sudantha and S. Sumathipala, "IoT Empowered Gesture Recognition System for Life Style Enhancement of Differently Abled People," *2018 International Conference on System Science and Engineering (ICSSE)*, 2018, pp. 1-5.

K. Warnakulasooriya, Y. P. Jayasuriya and B. H. Sudantha, "Generic IoT Framework for Environmental Sensing Research: Portable IoT Enabled Weather Station," *2018 International Conference on System Science and Engineering (ICSSE)*, 2018, pp. 1-5.

Y. P. Jayasuriya, C. S. Elvitigala, K. Wamakulasooriya, and B. Sudantha, "Low Cost and IoT-Based Greenhouse with Climate Monitoring and Controlling System for Tropical Countries," *2018 International Conference on System Science and Engineering (ICSSE)*, 2018.

K Warnakulassooriya, J Bandara, K Priyadarshana, “Low Cost Lab-scale Bioreactor for Optimization of Microbial Fermentation Processes,” *2nd International Conference in information technology research*, 2017. **[Best Paper Awarded]**

K Warnakulasooriya, BH Sudantha, K Sumathipala, “A Novel Approach of Color Detection Technique for the Enhancement of Robotic Navigation,” *18th International Symposium on Advanced Intelligent Systems (ISIS)*, 2017.

B. H. Sudantha, K. A. S. N. Sumathipala, C. Premachandra, K. Warnakulasooriya, C. S. Elvitigala and Y. P. Jayasuriya, "Adaptive navigation and motion planning for a mobile track robot," *2017 Joint 17th World Congress of International Fuzzy Systems Association and 9th International Conference on Soft Computing and Intelligent Systems (IFSA-SCIS)*, 2017.

B. H. Sudantha, K. Warnakulasooriya, Y. P. Jayasuriya, G. R. Ratnayaka, P. K. S Mahanama, E. J. Warusavitharana, S. N. Weerasinghe, “Open-source implementation of an integrated low-cost environmental monitoring system (EMS) for developing countries,” *Bhumi, The Planning Research Journal, 2018*.

K. Warnakulasooriya, J. Bandara and B. Sudantha, "Sensory Evaluation of Foods using Modern AI Techniques and Brain Wave Analysis," *2020 International Conference on Image Processing and Robotics (ICIPRoB)*, 2020.

ACADEMIC CONTRIBUTIONS

Scholarly Reviewer | Soft Computing **Nov 2024 - Present**
Optimization, Combinatorial Optimization, Intelligent Robotics, Multi-Robot Systems

Scholarly Reviewer | IJACSA Journal **Oct 2024 - Present**
Neural networks, Deep learning, Swarm Intelligence, Distributed Computing, intelligent robotics

Scholarly Reviewer | IEEE RA-L **May 2025 - Present**
Robotics, UAC systems, UAV control

TECHNICAL SKILLS

Technologies and frameworks
Robotics: ROS, ROS2, Gazebo, RViz, MavROS, RPC, PX4
Machine Learning: Pytorch, Scipy, Scilab, Mesa, SwarmX (Founder), matplotlib, and Seaborn
Front-End: React, Redux, NextJS, SCSS, AntUI
Mobile: React-Native, Expo
Backend: NodeJS, Express JS, Apollo, Springboot(Moderate), Anaconda, Jenv

Languages
Java, Python, Python3, C++, Javascript, Typescript, HTML/CSS, SCSS, SQL, Latex

Database
PostgreSQL, MySQL, MongoDB, DynamoDB, Redis

Continuous integration & delivery
Jenkins, GoCD, Git, Argo, Docker, Maven, Gradle

Testing, Performance & Quality
Junit, Cypress, Jest, Jmeter, SonarQube

Project Management and Documentation
Jira, Confluence, Github Pages

Communication
Slack, Outlook, Microsoft Teams

Editors, IDEs, and Tools
Visual Studio, Visual Studio Code, JetBrains IDE, Postman

Research Tools
Google Scholar, Semantic Scholar, Research Rabbit, Overleaf

AI Tools
Github Copilot, Microsoft Copilot, ChatGPT, codegen

Other
AWS, Datadog, Dynatrace, API central, Talend open studio, Optimizely, Google

PROJECTS

E2E Machine Learning Framework (github.com/kolithawarnakulasooriya/e2e-ml-flow)

- Developed a Software System framework for end-to-end ML system development using MLFlow and
- ZenML pipelines incorporate data ingestion and various statistical analysis techniques.
- Implemented data processing algorithms and data structures for efficient feature engineering and model training, using appropriate design patterns.
- Designed and implemented routing strategies for optimal data flow within the system, considering operating system constraints and resource management.
- Achieved 95% unit test code coverage, demonstrating proficiency in software development best practices and showcasing expertise in algorithms and data structures.

Drone-Controlling Framework

- Developed a ROS-based Python flight framework for heterogeneous commercial UAVs.

- Implemented adaptable ROS packages for Tello, Anafi, and Bebop UAVs, leveraging Linux development skills for controlling 15 UAVs using a hierarchical architecture.
- Designed and implemented a Communication Solution using custom Communication Protocols for real-time data streaming and command transmission between the ground station and UAVs.
- Integrated Tello camera video streams as fused sensor data for autonomous navigation and control, achieving a 95% success rate in autonomous waypoint navigation, and communication Protocol implementation.

SwarmX SI & MAS Simulator (github.com/kolithawarnakulasooriya/SwarmX)

- Developed a Java-based software system, a simulation tool for swarm intelligence optimization algorithms and multi-agent behaviors, incorporating low-latency data structures.
- Implemented 21 built-in SI algorithms and over 40 benchmark test cases using efficient data structures, ensuring optimal simulation performance.
- Designed and executed simulations, analyzing routing strategies and optimizing for low-latency performance within the software system.
- Achieved a 15% reduction in simulation runtime through optimized data structure implementation and efficient routing algorithms, demonstrating proficiency in simulation tool development, data structures, and software system design.

MineArc Service App | Freelanced (minearc.com/serviceapp)

- Developed a hybrid mobile application (Hybrid App Development) using React Native and React for iOS and Android platforms (iOS, Android, Mobile Application) to manage service terms.
- Built a web application (Web Application) front-end using React, integrating with a back-end system, leveraging web technologies (Web Technology) for Software Development.
- Implemented cross-platform mobile features (Mobile Application) for iOS and Android, enhancing user experience.
- Spearheaded the development of a responsive Web Application (Web Application) and Mobile Application (Mobile Application), improving cross-functional collaboration through code reviews, resulting in a 15% reduction in bug reports (Software Development).

Mockable Express

- Developed an open-source, in-memory mock API framework for testing machine learning and backend components, incorporating a user-friendly interactive UI.
- Designed and implemented core API functionalities using RESTful principles by leveraging core JavaScript programming to modify the Express.js node application server.

4ONSE Monitoring System

- Designed and deployed a solar-powered weather station network (34 stations) across Sri Lanka, incorporating 11 sensors and real-time data transmission to the ISTSOS server.
- Implemented data acquisition techniques using sensor fusion techniques and preprocessing algorithms using C++ programming, ensuring data integrity and efficient transmission.
- Developed and maintained a robust data pipeline using SQL for database management and visualization tools for data analysis and reporting in real-time.
- Achieved 99.5% data capture rate across all stations, demonstrating system design, Programming, and data management expertise.

OPEN SOURCE CONTRIBUTIONS

TransformerLab – LLM Payground **Jan 2025 – Present**

- Fixed two critical bugs related to the front-end and back-end
- Tested the Gen AI inference engine module.

EXTRA CURRICULAR

Formula 1 – SAE Hybrid Electric Competition Vehicle **Jan 2024 – May 2025**
University of South Alabama, Mobile, Alabama

- Achieved 3rd place in the hybrid category.
- Appointed as the computer team lead, designed and developed the electronic and computing system to collect, manage, and manipulate ECU, BMS, and motor control over the CAN network.
- Developed the FE application to display real-time data on the dashboard with bidirectional communication over the CAN network.

SERVICE AND LEADERSHIP

Future Leaders - Outbound Leadership Training Program
Ministry of Higher Education, Sri Lanka

HONORS AND AWARDS

Outstanding Reviewer Award – 2025
IJACSA Journal

SAE Formula1, 2024 – Second runners-up
SAE Team, University of South Alabama

Graduate Research Assistantship
School of Computing, University of South Alabama (2022)

Hall of Fame - 1st Runners Up
Sysco Delivery Application – SyscoLabs
Empowering customer experience with machine learning

Special Recognition
Sysco Shop – SyscoLabs

Best Paper Award
Low-cost lab-scale bioreactor for optimization of microbial fermentation processes
2nd International Conference in Information Technology Research